# Constellations of Prime Chains on a Logarithmic Spiral:
## Geometric Symmetries of Cunningham and Collatz-type Chains

*March 8, 2026 — Hiroshi Harada*

***Logarithmic Spiral Mapping:***

*radius* $r = \ln(n)$

$\theta = 2\pi \ln(n)$

$\Delta\theta \approx 2\pi \ln(k)$

***The Cunningham Triangle:***

$n \rightarrow 2n \pm 1$

***The Collatz Pentagram:***

$n \rightarrow (3n + 1) / 2$ *(applied only when the result is prime)*

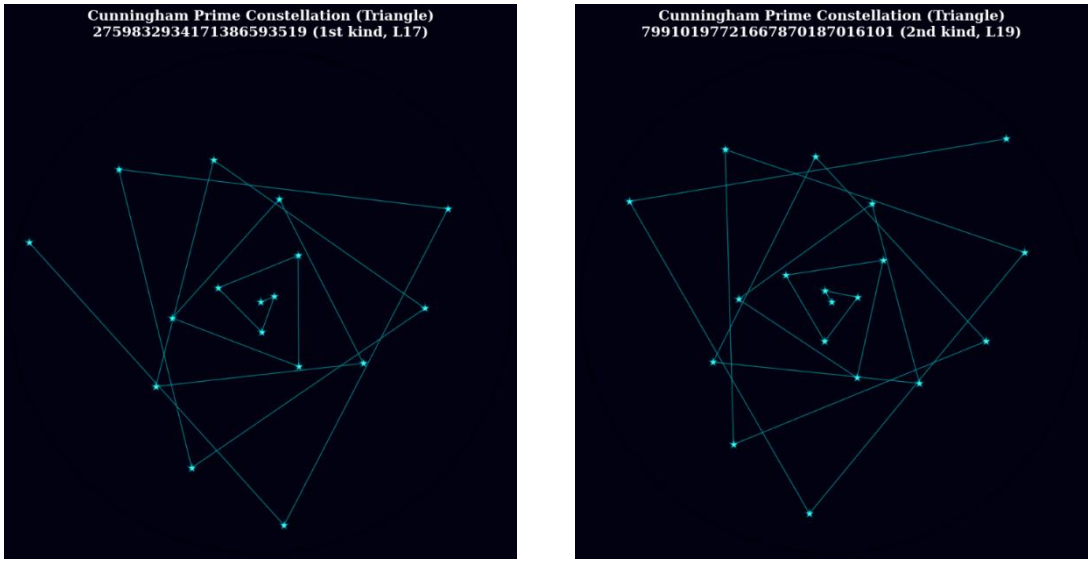


**Figure 1. The Cunningham Triangle.**
A logarithmic-spiral plot of a Cunningham prime chain. Because $\ln(2) \approx 2/3$, the sequence rotates nearly two full turns every three steps, producing a self-similar expanding triangular pattern.

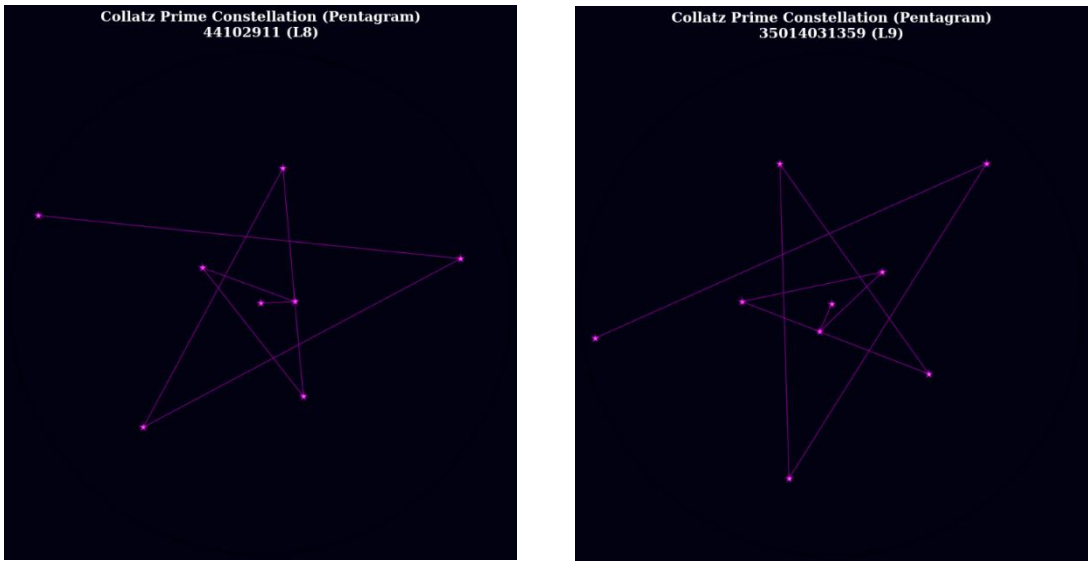


**Figure 2. The Collatz Pentagram.**
A logarithmic-spiral plot of a Collatz-type prime chain. Since $\ln(1.5) \approx 2/5$, the sequence rotates about two full turns every five steps, forming an intersecting, self-similar pentagram.